Proposed IMEX–ADER–DG Scheme for the One-Dimensional Low-Froude Shallow-Water Equations over Fixed Bathymetry

Discussion notes before implementation

October 2026

1 Purpose and references used in these notes

The goal is to propose a simple second-order IMEX–ADER–DG baseline for the one-dimensional low-Froude shallow-water equations over a fixed bathymetry $b = b(x)$, before proceeding to implementation.

Throughout these notes, the term *reference IMEX method* denotes the second-order IMEX Crank–Nicolson scheme of Kazolea–Lteif–Parisot (Computers & Fluids, 2025). The slow/fast splitting and the implicit treatment of the acoustic/free-surface terms are taken from the spirit of that paper.

For the ADER part, the previous ADER–DG notes remain the reference for the local space–time predictor, differential-transformation recurrences, polynomial product/reciprocal rules, predictor traces, space–time quadrature, and one-step DG correction. Those notes were written for a fully explicit ADER–DG method, but the same local ADER machinery is used here on the *explicit convective subsystem only*.

Some integral weak forms are omitted below when they do not add information, in order to keep the construction readable.

2 Low-Froude model and IMEX splitting

Let

$$\boxed{\eta = h + b, \qquad h = \eta - b, \qquad q = hu, \qquad u = \frac{q}{h},} \tag{1}$$

where $b = b(x)$ is fixed. The one-dimensional nondimensional system is

$$\boxed{\begin{aligned} \partial_t \eta + \partial_x q &= 0, \\ \partial_t q + \partial_x (qu) + \frac{h}{\varepsilon^2}\,\partial_x \eta &= 0, \end{aligned} \qquad \varepsilon = Fr.} \tag{2}$$

We separate

$$\boxed{\underbrace{\partial_x (qu)}_{\text{slow nonlinear convection: explicit}} \qquad \text{from} \qquad \underbrace{\left(\partial_x q,\ \frac{h}{\varepsilon^2}\partial_x \eta\right)}_{\text{fast acoustic/free-surface part: implicit}}.} \tag{3}$$

The reason for treating the acoustic block implicitly is the same as in the reference IMEX method: the fast gravity-wave scale is $O(Fr^{-1})$, and we do not want the time step to be restricted by this scale.

3 Relation with the reference IMEX method

In one space dimension, the reference second-order IMEX construction has the structure

$$\mathcal{F}^{n+\frac{1}{2}} = q^n - \frac{\Delta t}{2}\,\partial_x\Big(\widehat{q}^{\,n+\frac{1}{2}}\widehat{u}^{\,n+\frac{1}{2}}\Big) - \frac{\Delta t}{2\varepsilon^2}\,h^{n+\frac{1}{2}}\partial_x\eta^{n+\frac{1}{2}}, \tag{4}$$

$$q^{n+1} = q^n - \Delta t\,\partial_x\Big(\widehat{q}^{\,n+\frac{1}{2}}\widehat{u}^{\,n+\frac{1}{2}}\Big) - \frac{\Delta t}{\varepsilon^2}\,h^{n+\frac{1}{2}}\partial_x\eta^{n+\frac{1}{2}}, \tag{5}$$

where

$$\widehat{\psi}^{\,n+\frac{1}{2}} = \frac{3}{2}\psi^n - \frac{1}{2}\psi^{n-1}, \qquad \eta^{n+\frac{1}{2}} = \frac{\eta^n + \eta^{n+1}}{2}. \tag{6}$$

Thus convection is explicit through AB2 extrapolation, while the free-surface/gravity contribution is implicit through midpoint quantities.

The proposed method keeps this same IMEX separation. The main change is

$$\boxed{\partial_x\Big(\widehat{q}^{\,n+\frac{1}{2}}\widehat{u}^{\,n+\frac{1}{2}}\Big) \quad \longrightarrow \quad C^{n+\frac{1}{2}}_{h,\mathrm{ADER}},} \tag{7}$$

where $C^{n+1/2}_{h,\mathrm{ADER}}$ is obtained from a local ADER space–time predictor.

A provisional semi-implicit midpoint state is introduced before ADER because the local ADER predictor should be centered around an acoustically corrected approximation of the solution at $t^{n+1/2}$, rather than around the old state U^n_h.

	Reference IMEX method	Proposed IMEX–ADER–DG method
Fast acoustic part	Implicit midpoint / Crank–Nicolson	Same implicit midpoint philosophy
Nonlinear convection	Explicit AB2 extrapolation	Explicit local ADER predictor and time average
Spatial discretization	Cell-centered finite volume	DG; compatible Local DG derivatives for the acoustic part
Midpoint state before explicit convection	Not required by AB2	Provisional semi-implicit midpoint state $\bar{U}_h$
Pressure coefficient	Midpoint depth	Known positive lagged depth: h^n_h in the provisional solve and $\bar{h}_h$ in the final solve

Important design rule.

$$\boxed{\text{No } \varepsilon^{-2} \text{ acoustic term is inserted into the local ADER recurrence.}} \tag{8}$$

If an ε^{-2} term enters the explicit Taylor/DT coefficients, the local predictor inherits the stiff gravity-wave scale and can reintroduce the low-Froude stability restriction that the IMEX splitting is meant to remove. Therefore the stiff acoustic contribution is kept entirely inside the global implicit solves, while ADER evolves only the slow convective part.

4 Minimal DG and Local DG notation

Let

$$I_j = [x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}], \qquad V^k_h = \{v_h :\ v_h|_{I_j} \in \mathbb{P}^k(I_j)\}.$$

At time t^n,

$$U_h^n = (\eta_h^n, q_h^n)^T \in [V_h^k]^2, \qquad h_h^n = \eta_h^n - b > 0.$$

At each interior interface, v^- and v^+ denote the traces from the left and right cells. At a physical boundary, the missing exterior trace is supplied by the boundary condition of the problem.

4.1 Compatible Local DG derivatives for the acoustic part

We use a right-trace DG gradient $\mathcal{G}_h^+$ and the opposite left-trace DG divergence $\mathcal{D}_h^-$. For every test function in the local polynomial space,

$$\boxed{\int_{I_j} (\mathcal{G}_h^+ v_h)\varphi_h \, dx = -\int_{I_j} v_h \, \partial_x \varphi_h \, dx + v^+_{j+\frac{1}{2}} \varphi^-_{j+\frac{1}{2}} - v^+_{j-\frac{1}{2}} \varphi^+_{j-\frac{1}{2}},} \tag{9}$$

$$\boxed{\int_{I_j} (\mathcal{D}_h^- r_h)\psi_h \, dx = -\int_{I_j} r_h \, \partial_x \psi_h \, dx + r^-_{j+\frac{1}{2}} \psi^-_{j+\frac{1}{2}} - r^-_{j-\frac{1}{2}} \psi^+_{j-\frac{1}{2}}.} \tag{10}$$

The opposite traces are chosen because they make the two discrete derivatives adjoint-compatible. With a positive depth coefficient, this gives a coercive acoustic operator and a uniquely solvable linear implicit problem.

Products such as $a_h \mathcal{G}_h^+ v_h$ are understood in the DG weak sense. The corresponding integral form is omitted below unless needed explicitly.

4.2 Explicit convective DG residual

For

$$f_c(U_h) = \frac{q_h^2}{h_h} = q_h u_h, \qquad h_h = \eta_h - b,$$

define $\mathcal{C}_h(U_h)$ by the usual DG residual. Cellwise,

$$\boxed{\begin{aligned}\int_{I_j} \mathcal{C}_h(U_h)\psi_h \, dx = &-\int_{I_j} f_c(U_h) \, \partial_x \psi_h \, dx \\ &+ \widehat{f}_{c,j+\frac{1}{2}} \psi^-_{j+\frac{1}{2}} - \widehat{f}_{c,j-\frac{1}{2}} \psi^+_{j-\frac{1}{2}}.\end{aligned}} \tag{11}$$

A simple baseline choice is the Rusanov flux

$$\boxed{\widehat{f}_c(U^-, U^+) = \frac{1}{2}\left(\frac{(q^-)^2}{h^-} + \frac{(q^+)^2}{h^+}\right) - \frac{\alpha_c}{2}(q^+ - q^-), \qquad \alpha_c = \max(2|u^-|, 2|u^+|).} \tag{12}$$

Only the convective speed is used; the gravity-wave speed $\sqrt{h}/Fr$ is deliberately excluded from the explicit numerical viscosity.

5 Proposed second-order IMEX–ADER–DG time step

Assume $U_h^n = (\eta_h^n, q_h^n)^T$ is known.

5.1 Step A: provisional semi-implicit midpoint state

First compute

$$\boxed{C_h^n = \mathcal{C}_h(U_h^n).} \tag{13}$$

Then define $\bar{U}_h = (\bar{\eta}_h, \bar{q}_h)^T$ by

$$\boxed{\bar{\eta}_h = \eta_h^n - \frac{\Delta t}{2} \mathcal{D}_h^- \bar{q}_h,} \tag{14}$$

$$\boxed{\bar{q}_h = q_h^n - \frac{\Delta t}{2} C_h^n - \frac{\Delta t}{2\varepsilon^2}\, h_h^n \mathcal{G}_h^+ \bar{\eta}_h.} \tag{15}$$

The convection is explicit through C_h^n, while the mass/acoustic coupling is implicit. The coefficient h_h^n is already known and positive.

Eliminating $\bar{q}_h$ gives the linear implicit problem

$$\boxed{\bar{\eta}_h - \frac{\Delta t^2}{4\varepsilon^2} \mathcal{D}_h^- \left(h_h^n \mathcal{G}_h^+ \bar{\eta}_h \right) = \eta_h^n - \frac{\Delta t}{2} \mathcal{D}_h^- q_h^n + \frac{\Delta t^2}{4} \mathcal{D}_h^- C_h^n.} \tag{16}$$

After solving it,

$$\boxed{\bar{h}_h = \bar{\eta}_h - b, \qquad \bar{u}_h = \frac{\bar{q}_h}{\bar{h}_h}.} \tag{17}$$

Role of this stage. It provides the ADER predictor with an approximation of the full midpoint state, while the stiff acoustic term remains in a global implicit solve.

5.2 Step B: local convective-only ADER predictor

Set

$$t^{n+\frac{1}{2}} = \frac{t^n + t^{n+1}}{2}, \qquad \tau = t - t^{n+\frac{1}{2}}, \qquad -\frac{\Delta t}{2} \le \tau \le \frac{\Delta t}{2}.$$

In each cell, solve only

$$\boxed{\partial_\tau U_{\tau,j}^E + \partial_x F^E(U_{\tau,j}^E) = 0, \qquad F^E(U) = \begin{pmatrix} 0 \\ qu \end{pmatrix},} \tag{18}$$

with

$$\boxed{U_{\tau,j}^E(x,0) = \bar{U}_{h,j}(x).} \tag{19}$$

Since the reference IMEX time discretization is second order, we use the corresponding second-order ADER time representation,

$$U_{\tau,j}^E(x,\tau) = \sum_{k_x=0}^{k} \widetilde{U}_j^E(k_x,0)(x-x_j)^{k_x} + \tau \sum_{k_x=0}^{k-1} \widetilde{U}_j^E(k_x,1)(x-x_j)^{k_x}, \tag{20}$$

with

$$\boxed{\widetilde{U}_j^E(k_x,1) = -(k_x+1)\widetilde{F}_j^E(k_x+1,0).} \tag{21}$$

All ADER algebra needed to evaluate the coefficients is the same as in the previous ADER–DG notes.

If

$$\bar{\eta}_{h,j}(x) = \sum_{k_x=0}^{k} \bar{\eta}_{j,k_x}(x-x_j)^{k_x},$$

then

$$\boxed{\widetilde{\eta}_j^E(k_x,0) = \bar{\eta}_{j,k_x}, \qquad \widetilde{\eta}_j^E(k_x,k_t) = 0 \quad (k_t \ge 1),} \tag{22}$$

so

$$\boxed{\eta_{\tau,j}^E(x,\tau) = \bar{\eta}_{h,j}(x).} \tag{23}$$

The discharge evolves through

$$\boxed{\widetilde{q}_j^E(k_x,1) = -(k_x+1)\widetilde{(qu)}_j^E(k_x+1,0),} \tag{24}$$

with $h_\tau^E = \bar{h}_h$ and $u_\tau^E = q_\tau^E / \bar{h}_h$.

Interpretation. The free surface is not set to zero; during this local convective subproblem it is frozen at the acoustically corrected midpoint polynomial $\bar{\eta}_h$. The only evolving ADER component is the explicitly treated convective discharge.

5.3 Step C: ADER time-averaged convection

The explicit contribution used in the final update is

$$\boxed{C_{h,\mathrm{ADER}}^{n+\frac{1}{2}} = \frac{1}{\Delta t}\int_{-\Delta t/2}^{\Delta t/2} \mathcal{C}_h\big(U_\tau^E(\tau)\big)\, d\tau.} \tag{25}$$

At the interfaces, the same nonstiff numerical flux (for example, the Rusanov flux (12)) is evaluated using the left and right ADER predictor traces and integrated in time, exactly in the spirit of the previous ADER–DG notes.

Equation (25) is the direct replacement of the AB2 explicit convective approximation in the reference IMEX method.

5.4 Step D: final implicit Crank–Nicolson acoustic correction

Define

$$\boxed{m_h = \eta_h^{n+\frac{1}{2}} = \frac{\eta_h^n + \eta_h^{n+1}}{2}.} \tag{26}$$

Use the already computed midpoint depth $\bar{h}_h$ as the known pressure coefficient:

$$\boxed{q_h^{n+1} = q_h^n - \Delta t\, C_{h,\mathrm{ADER}}^{n+\frac{1}{2}} - \frac{\Delta t}{\varepsilon^2}\,\bar{h}_h \mathcal{G}_h^+ m_h.} \tag{27}$$

Define

$$\boxed{\mathcal{F}_h^{n+\frac{1}{2}} = q_h^n - \frac{\Delta t}{2} C_{h,\mathrm{ADER}}^{n+\frac{1}{2}} - \frac{\Delta t}{2\varepsilon^2}\,\bar{h}_h \mathcal{G}_h^+ m_h,} \tag{28}$$

and update

$$\boxed{\eta_h^{n+1} = \eta_h^n - \Delta t\, \mathcal{D}_h^- \mathcal{F}_h^{n+\frac{1}{2}}.} \tag{29}$$

As in the reference IMEX structure,

$$\boxed{\mathcal{F}_h^{n+\frac{1}{2}} = \frac{q_h^n + q_h^{n+1}}{2}.} \tag{30}$$

For implementation, solve directly for m_h:

$$\boxed{m_h - \frac{\Delta t^2}{4\varepsilon^2}\mathcal{D}_h^-\big(\bar{h}_h \mathcal{G}_h^+ m_h\big) = \eta_h^n - \frac{\Delta t}{2}\mathcal{D}_h^- q_h^n + \frac{\Delta t^2}{4}\mathcal{D}_h^- C_{h,\mathrm{ADER}}^{n+\frac{1}{2}}.} \tag{31}$$

Then $\eta_h^{n+1} = 2m_h - \eta_h^n$ and q_h^{n+1} follows from (27).

6 Order and practical implementation choices

Temporal order. The present baseline is second order in time: the implicit part uses the second-order midpoint/Crank–Nicolson construction, and the ADER predictor is also truncated consistently at second order in time. Increasing the ADER time order alone would not increase the overall time order. To construct a higher-order method, both the implicit IMEX discretization and the explicit ADER discretization should be upgraded consistently.

Convective numerical flux. For the first implementation, a Rusanov flux based only on the convective eigenvalue $2u$ is a simple robust choice. The same flux should be used for C_h^n and for the time-dependent ADER predictor traces in $C_{h,\mathrm{ADER}}^{n+1/2}$.

Implicit linear solver. Both acoustic equations (16) and (31) are linear because the positive depth coefficient is frozen before each solve. For a first one-dimensional implementation, a sparse direct solver is the simplest validation choice. Since the compatible Local DG pair gives a symmetric/coercive acoustic operator under positive depth, a preconditioned conjugate-gradient solver is a natural scalable alternative. No Newton iteration is needed for this frozen-coefficient baseline.

7 Properties that should be established for this baseline

Under the assumptions used in the accompanying analysis (positive depth, well-prepared low-Froude data, fixed spatial discretization, and a bounded regular ADER convective map), the following properties have been established:

1. **Linear coercive acoustic solves.** With the alternating right/left Local DG pair and a positive frozen depth coefficient, the two implicit acoustic problems are coercive and uniquely solvable.

2. **Asymptotic-preserving behavior at fixed discretization.** As $Fr \to 0$ with Δt independent of Fr, the discrete solution converges to the corresponding discrete lake-limit system: the leading free surface is spatially constant and the leading discharge satisfies the discrete divergence constraint.

3. $O(Fr^2)$ **dynamic free-surface control.** The coercive acoustic solve gives a uniform bound on the scaled mean-zero free-surface component, which is the key estimate behind the fixed-discretization AP result.

4. **Second-order baseline.** For smooth positive-depth solutions, lagging the pressure coefficient at h_h^n and then $\bar{h}_h$ is consistent with the second-order midpoint construction.

5. **Exact midpoint mass-flux identity.** Equation (30) holds algebraically.

A separate linearized P^2 Fourier study showed stable low-Froude behavior for the tested smooth configurations under a velocity-based CFL independent of Fr; this is a numerical verification, not a nonlinear stability theorem.

The following points are intentionally left for later discussion/analysis: a nonlinear mechanical-energy estimate for arbitrary finite Fr, positivity/robustness for strong shocks and wet/dry states, well balancing, shock limiting, and the extension to the shallow-water–Exner system.

8 Compact view of one time step

$$
\begin{array}{|ll|}
\hline
U_h^n \longrightarrow \bar{U}_h & \text{(semi-implicit acoustic midpoint)} \\
\quad \longrightarrow \text{local ADER predictor} & \text{(explicit convection only)} \\
\quad \longrightarrow C_{h,\mathrm{ADER}}^{n+1/2} & \text{(time-averaged explicit residual)} \\
\quad \longrightarrow U_h^{n+1} & \text{(final implicit CN acoustic correction).} \\
\hline
\end{array}
\tag{32}
$$

The central idea is therefore simple: retain the low-Froude implicit acoustic treatment of the reference IMEX method, replace its explicit AB2 convection by the ADER–DG explicit machinery, and keep the stiff factor Fr^{-2} completely outside the local ADER recurrence.

9 Points for discussion before implementation

The construction above is the baseline proposed for the first code. In particular, feedback would be useful on the following choices:

1. whether the alternating right/left Local DG pair is the most natural DG realization of the centered acoustic treatment used in the reference IMEX method;
2. whether lagging the positive pressure coefficient at h_h^n and then $\bar{h}_h$ is preferable to retaining a nonlinear midpoint coefficient;
3. whether the nonstiff Rusanov flux with $\alpha_c = \max(2|u^-|, 2|u^+|)$ is a suitable first choice for the explicit convective residual;
4. and, for a later higher-order extension, which higher-order IMEX treatment should be paired with a correspondingly higher-order ADER predictor.

References. M. Kazolea, R. Lteif and M. Parisot, "An efficient second order ImEx scheme for the shallow water model in low Froude regime," *Computers & Fluids* 299 (2025), 106699.
Previous working notes: *ADER DG EXNER Ralph Ali Sacha*, September 2026.